Python Practice: Lists

Part 1 of 5 | 10 questions | Beginner to Expert

RULES: No for/while loops, no comprehensions, no recursion, no map/filter/reduce, and no hardcoded answers. Use indexing, slicing, operators, built-in functions, and data structure methods.

LEVEL: BEGINNER

Q1. Accessing List Elements

```
numbers = [10, 20, 30, 40, 50, 60, 70]
```

Extract the first, middle, and last elements without modifying the original list.

Q2. List Slicing

```
numbers = [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
```

Create three new lists: first five elements, last four elements, and every second element in reverse order.

Q3. List Modification

```
fruits = ["apple", "banana", "cherry", "mango"]
```

Replace banana with orange, add grapes at index 2, and remove cherry.

LEVEL: INTERMEDIATE

Q4. Reverse Without reverse()

```
numbers = [12, 45, 23, 67, 89, 34]
```

Reverse using slicing only; preserve the original list.

Q5. Second Largest Unique Number

```
numbers = [45, 12, 89, 34, 89, 67, 45, 23]
```

Find the second-largest unique number. Expected: 67.

Q6. Rotate a List

```
numbers = [10, 20, 30, 40, 50, 60]
k = 2
```

Rotate the list right by k using slicing. Expected: [50, 60, 10, 20, 30, 40].

Q7. Remove Duplicates, Preserve Order

```
numbers = [10, 20, 10, 30, 20, 40, 30, 50]
```

Remove duplicates while preserving first-occurrence order.

LEVEL: ADVANCED

Q8. Flatten a Nested List

```
data = [[10, 20], [30, 40], [50, 60]]
```

Flatten into [10, 20, 30, 40, 50, 60] using list operations.

Q9. Extract Data from a Matrix

```
matrix = [
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
]
```

Extract main diagonal, secondary diagonal, and middle column; calculate the sum of each.

Q10. List Transformation Challenge

```
data = [15, 5, 25, 10, 35, 20, 30]
```

Build a list of the three largest unique values ascending, then two smallest unique values descending. Expected: [25, 30, 35, 10, 5].